

A Thyristor-controlled Superconducting Current Limiting Module on Mitigation of Commutation Failure in LCC-Based HVDC Systems

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Abstract—With more applications and larger transmission capacity of LCC-HVDC transmission systems, commutation failure has attracted more attention. In view of this problem, existing methods pay less attention to commutation failures caused by severe short-circuit faults on AC or DC side. In this paper a thyristor-controlled superconducting current limiting module is proposed based on resistive superconducting current limiting units. The operating principles are presented followed by detailed analysis for commutation under AC short-circuit faults and DC short-circuit faults. The performance of the proposed method is validated by simulation results in MATLAB/Simulink. The simulation results show that the proposed method can effectively prevent the first commutation failure caused by severe short circuit fault.

Keywords—Commutation failure, Superconducting coil, LCC-based HVDC, Serious fault

I. INTRODUCTION

High voltage direct current (HVDC) transmission based on line-commutated converter (LCC) technology has been widely utilized around the world for power transmission. However, some well-known problems associated with LCC-HVDC still exist today which limits the further application of this technology. The most notable one is commutation failure. Commuting failure is one of the common faults of thyristor inverters [1][2]. Although the frequency of commutation failure varies from year to year, there is an obvious upward trend on the whole. The main causes of commutation failure include AC short circuit fault, the increase of non-fault pole current due to electromagnetic coupling between DC lines after the single pole to ground short circuit fault in the bipolar DC system, and the valve fault which is mainly caused by triggering abnormality. 86% of commutation failures were due to AC voltage sags caused by AC short circuit faults, of which 41% were serious faults with AC voltage sags of more than 50% or even 100% before commutation failure [3].

The commutation failure can cause a temporary cessation of power transfer and overheating of the valves. A sudden increase of power transfer between adjacent AC transmission lines can also happen which might affect system transient stability [4]. The first commutation failure is one of the

inducements of the cascading failures in the power system [5]. This puts forward higher requirements for the mitigation strategy of commutation failure: the strategy should be taken as an important part of the measures to prevent the cascading failures in the power system. It can reduce the probability of commutation failures and the impact on the power system after the occurrence of commutation failures. The key of reducing the impact is to limit the DC short-circuit current, to prevent the occurrence of continuous commutation failure, which can avoid the abnormal action of the DC and AC lines.

In recent years, with the breakthrough of superconducting materials, the method of using superconducting fault current limiter (SFCL) to suppress commutation failure has gained some attention. In addition to the advantages of resistors, SFCLs can be kept in series in the system line during steady-state operation. There is no need to connect full-control devices in parallel. SFCLs generate resistance after a fault occurs, so the operating time does not depend on the fault detection. Therefore, it has certain advantages to use the superconducting current limiter-based suppression method as an auxiliary method to adjust the inverter strategy.

This paper is organized as follows. In Section 2, the proposed method on mitigation of commutation failure based on SFCLs is described. Section 3 explains the principles of SFCL and operation of the proposed method. In Section 4, analytical derivations of the commutation period with the SFCL are presented. In Section 5, the effectiveness of the proposed method is validated by various simulations in PSCAD. Finally, Section 6 concludes the paper.

II. THE INVERTER WITH SFCLS

Fig. 1 shows the proposed configuration where VTB and VTF are thyristor valves. SupCoil is superconducting coils.

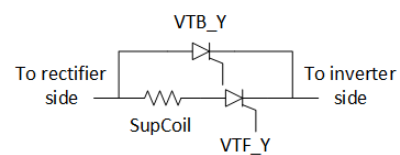


Fig. 1. The proposed superconducting module configuration.

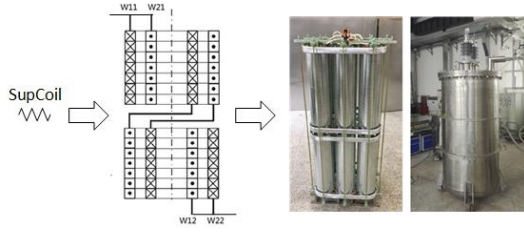


Fig. 2. Superconducting coil.

There are superconducting coils in superconducting modules shown in Fig. 1. Each superconducting coil can be realized by several series and parallel connected sub-coils to achieve higher insertion resistance and higher critical current. In the following modeling, single superconducting coil is assumed for the sake of demonstration of the approach.

The superconducting coil shown in Fig. 2 is constructed by a solenoidal configuration. The superconducting coil consists of 2 inner and 2 outer cylinders. The 4 cylinders are arranged coaxially. 1 outer cylinder and 1 inner cylinder form a group. The inner cylinder in the upper group is connected in series with the outer cylinder in the lower group and the outer cylinder in the upper group is connected in series with the inner cylinder in the lower group to ensure that the same amount of current through the inner and outer cylinders in each group. The inner and outer cylinders of each group are spiraled in opposite directions to ensure that the magnetic fields cancel each other out so that the superconducting coils are resistive. The measured leakage inductance of a superconducting coil is only $3.7 \mu H$ when the difference in radius between the inner and outer cylinders is 10 mm [7]. The skeleton of the cylinder is made of epoxy resin and the outer layer of the skeleton is slotted. Superconducting tapes are wound around it. In this paper, an $Y_2O_3 \cdot BaO \cdot CuO$ (YBCO) tape called AMSC8602 is chosen. Several superconducting coils are assembled in series and parallel and then placed into a Dewar and connected to refrigeration system to ensure that superconducting coils are immersed in liquid nitrogen.

Superconducting coils exhibit two states: the superconducting state and the normal state, which is shown in Fig. 3. When the current through the superconducting coil is less than its critical current, the coil is in the superconducting state and its resistance is zero. When the current through the superconducting coil is greater than its critical current, the coil quenches. Then the coil is in the normal state and has resistance. When the current is cut off and after cooling, the superconducting coil returns to the superconducting state. In the superconducting module shown in Fig. 1, the superconducting coil will be inserted when VT2F_Y and VT5F_Y are switched on, meanwhile, VT2_YIN and VT5_YIN are switched off. The superconducting coil will be bypassed when VT2_YIN and VT5_YIN are switched on, meanwhile, VT2F_Y and VT5F_Y are switched off.

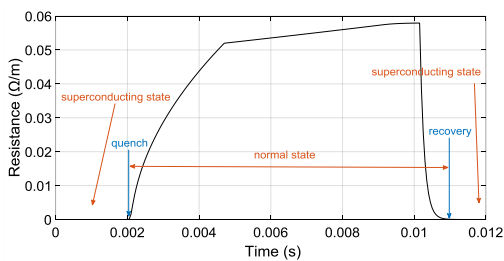


Fig. 3. Resistance of superconducting coils under an impact.

III. CONTROL STRATEGIES

A. Commutation failure

The essence of commutation failures is that the valve in the outgoing phase fails to be switched off because its current does not drop to zero in a required time. Theoretical analysis of commutation failures is presented in [1], which concludes that the inverter AC voltage magnitude reduction is the most dominant factor for commutation failure under both single-phase and three-phase fault. Voltage phase angle shift is also the reason for the onset of commutation failure. Another important conclusion is that the DC current increase caused by single-pole fault and electromagnetic coupling between DC lines [8]. Valve faults can also cause commutation failures. It is due to fact that the control system will not be able to eliminate commutation failures caused by serious faults [6]. However if the aim is to eliminate commutation failures rather than reduce its probability, the most direct way is to reduce the current of the outgoing phase. In this paper, SFCLs are adopted to eliminate the commutation failures under zero impedance single-phase and three-phase faults at inverter AC bus, zero impedance single-pole faults at DC lines and valve faults. The usefulness of the method then largely depends on the control strategy and selection of superconducting coil resistance which are described in following sections. In a commutation, the outgoing phase and the incoming phase are shown in Fig. 4.

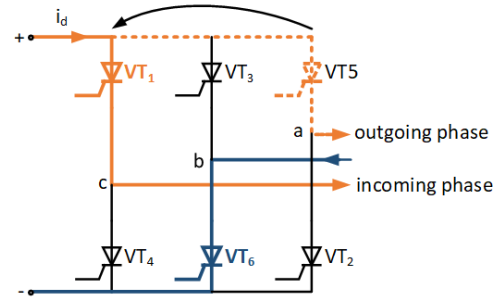


Fig. 4. The outgoing phase and the incoming phase in a commutation.

B. Operating principles

The control strategy is called 'flow' & 'bypass' method. If faults occur in the outgoing phase, superconducting modules will keep in series in the circuit. If faults occur in the incoming phase, superconducting modules will be bypassed.

Through the parallel connection of several superconducting coils, the critical current of the superconducting module is appropriately higher than the rated current of the line where it is located. The superconducting module usually shows zero resistance, which does not affect the normal operation of the line and the inverter. When a serious fault occurs, the line current rises rapidly and the superconducting coil automatically quenches. The superconducting module maintains the flow or bypass state according to the control logic. Refer to Section 4 for the definition of serious faults and minimum limiting resistance. The action process of the superconducting module can be divided into 4 stages.

Stage I: Before the short-circuit fault occurs, the line is operating normally. The superconducting module is in the flow state. At this time, the line current is lower than the critical current of the superconducting module. The superconducting coil is in the superconducting state, which

shows zero resistance. The voltage sensor of the superconducting module monitors the voltage at both ends of superconducting coils to determine whether it quenches.

Stage II: When a short circuit fault occurs, the line current exceeds the critical current of the superconducting module. The superconducting coil quenches. The voltage at both ends of the superconducting module exceeds the threshold, then the superconducting module controller judges the line type and fault type. If the line type is AC side incoming phase or the fault type is non-serious fault, the superconducting module should be in the bypass state. Otherwise, the superconducting module should be in the flow state.

Stage III: If the superconducting module is in the flow state after Stage II, the controller judges the line type and the fault type again, i.e., repeats Stage II.

Stage IV: If the superconducting module is in the bypass state after Stage II, it is determined that the action process is over when the voltage sensor of the superconducting module detects that the voltage at both ends of superconducting coils drops below the threshold. The superconducting module should be in the flow state and return to Stage I.

The control logic of superconducting modules is shown in Fig. 5.

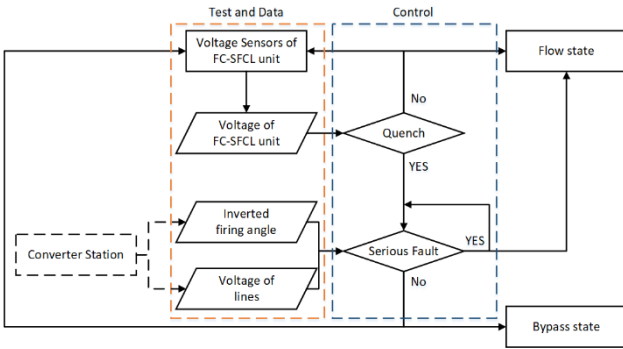


Fig. 5. Operating principles of superconducting modules.

The advantage of SFCLs is that in the steady state, superconducting coils are in the superconducting state, which does not affect the operation of the system. The superconducting module can always operate in the flow state when there is no fault. When there is a short-circuit fault in the line, the superconducting module quenches because the operating current exceeds the critical current, which limits the short-circuit current. After the superconducting module in the bypass state returns to the superconducting state, a trigger signal is sent to the thyristor and the superconducting module is re-inserted into the line and returns to the flow state.

Fig. 6 shows the control logic for switching from the bypass state to the flow state. Measure the voltage across superconducting coils in the bypass state. If the measured value is lower than the threshold value, it is considered that the superconducting module has completed the quench recovery. The state of the module changes from the bypass state to the flow state. If the voltage measurement value is higher than the threshold value, it is considered that coils is still in the normal state, the superconducting module remains in the bypass state and continues to measure its voltage.

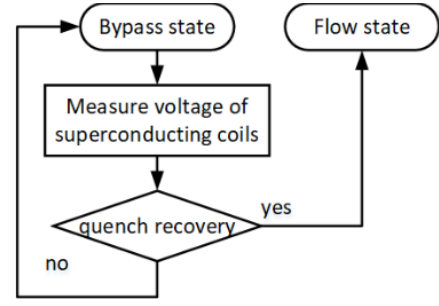


Fig. 6. Control logic from bypass state to flow state.

The superconducting module is in flow state before the fault occurs and at the initial stage of the fault. Only when the fault occurs and the superconducting module quenches, the controller will start to judge the fault type, then bypass the superconducting module which should not be in the flow state and has quenched. So requirements for the fault judgment time are not too high.

There are two scenarios in which the superconducting module needs to be bypassed in time: the fault line in the single-pole grounding short-circuit fault of the bipolar HVDC transmission system and the incoming phase in the AC short-circuit fault. Except for these two situations, keeping the superconducting module in the flow state is beneficial to suppress the commutation failure. Therefore, the key point of the control logic is how to transfer the superconducting module which has quenched but not needed from the flow state to the bypass state during the fault. In order to suppress commutation failures effectively, it should be prudent to judge that the superconducting module changes from the flow state to the bypass state. The judgment principle of this process is that assuming that there is no superconducting module, whether the HVDC transmission system will occur commutation failure, that is, to judge according to

$$\cos \gamma_{min} > \cos \beta + \frac{2\omega L i_d}{u_{\Delta f}} \quad 1$$

where γ_{min} , ω , L are constants, i_d , $u_{\Delta f}$ represent the instantaneous values of the DC current and the commutation voltage, respectively.

Fig. 7 shows the control logic from the flow state to the bypass state. The blue and red dotted boxes are the specific control processes of the superconducting bridge module and the superconducting DC module, respectively. The other parts are the control processes common to all modules. The method of judging whether the quench bridge module is on the incoming phase is to judge whether the latest trigger signal before the fault is sent to the thyristor of the phase. If it is sent to the phase, the corresponding phase is the incoming phase or is not in the commutation process. It is ensured that the phase must not be the outgoing phase and bypassing the superconducting module will not have a negative impact on suppressing the commutation failure. If the last trigger signal is not sent to this phase, then this phase must not be the incoming phase. So maintaining the flow state of the superconducting module will not deteriorate the commutation process.

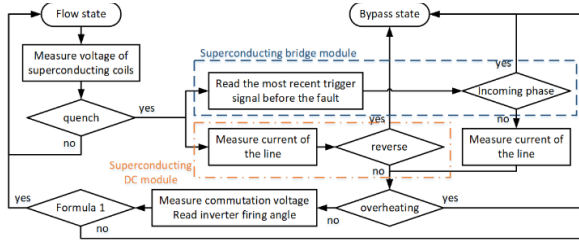


Fig. 7. Control logic from flow state to bypass state.

The method of judging whether the quench DC module is on the fault line is to judge whether the current of the DC line where it is located is reversed. In a DC line, only the inverter side current on the fault line will reverse in the event of a single-pole grounding short-circuit fault, while the non-fault line inverter side current will not reverse under no fault or any fault condition. This part of the judgment can achieve fast action.

After this part of the judgment, it is necessary to calculate the resistance of superconducting coils to determine whether superconducting coils is overheated and bypass them if it is damaged by overheating. Finally, measure the commutation voltage, read the inverter firing angle and judge whether the superconducting module needs to be kept in the flow state by formula 1.

IV. THEORETICAL ANALYSIS AND PARAMETER SELECTION

When the power system operates normally, the commutation circuit is constant. The derivation of commutation only considers inductances of lines and transformers. Due to factors such as the commutation voltage drop or the change of the DC current, number of parameters which can affect commutation increases and even the structure of the commutation circuit is changed under faults. In addition, the influence of superconducting coil on the commutation circuit can't be negligible. In this section, formulas for the extinction angle of commutation with SFCL under 2 typical faults are derived: AC short-circuit fault and single-pole grounding short-circuit (SPG) fault in the bipolar DC system. In order to avoid duplication, the formula under AC short-circuit fault is deduced here in detail and the formulas under the other fault is given directly.

Under the AC short-circuit fault, voltage of the fault phase drops. The SFCL on the fault phase acts as a resistor. Take the commutation where the inverter valve VT_2 commutating to VT_4 as an example, the commutation circuit is shown in Fig. 8. R_{si} represents resistance of the SFCL on the i -th phase and its effective range is $R_{si} \geq 0$. $R_{si} = 0$ means that the SFCL on the i -th phase is not operating. k_{fi} represents the unit value of the voltage of the i -th phase under a short-circuit fault and its effective range is $0 \leq k_{fi} \leq 1$. $k_{fi} = 0$ means that a zero-resistance grounding short-circuit fault occurs at the primary winding of the converter transformer on the i -th phase. $k_{fi} = 1$ means that there is no short-circuit fault on the i -th phase. In a commutation (about 0.6 ms), R_{si} and k_{fi} can be approximated as constants.

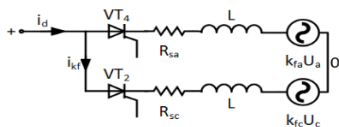


Fig. 8. The commutation circuit during an AC short-circuit fault.

During a normal commutation, the DC current i_d is regarded as a constant. According to Kirchhoff voltage laws, the first-order non-homogeneous linear differential equation of the commutation current i_{kf} can be calculated by

$$\frac{di_{kf}}{dt} + \frac{R_s}{2L} i_{kf} = \frac{R_{sa} i_d + U_{\Delta f}}{2L} \quad 2$$

where R_s and $U_{\Delta f}$ can be written as

$$R_s = R_{sc} + R_{sa} \quad 3$$

$$U_{\Delta f} = k_{fa} U_a - k_{fc} U_c \quad 4$$

where R_s represents the total resistance of SFCLs in the circuit. $U_{\Delta f}$ represents the commutation voltage under the AC short-circuit fault. By expanding U_a and U_c , formula 4 can be written as

$$U_{\Delta f} = k_f U \sin(\omega t - \varphi_k) \quad 5$$

where k_f and φ_k can be written as

$$k_f = \sqrt{2(k_{fa}^2 + k_{fc}^2 + k_{fa} k_{fc})} \quad 6$$

$$\varphi_k = \arctan \frac{\sqrt{3} k_{fc}}{2 k_{fa} + k_{fc}} \quad 7$$

where k_f and φ_k represent the unit amplitude and phase of the commutation voltage under the AC short-circuit fault, respectively. φ_k characterizes the zero-crossing point of $U_{\Delta f}$. When an AC asymmetrical fault such as a single-line grounding short-circuit (SLG) fault occurs as severity of the fault changes, φ_k will also change. Take an SLG fault on phase C as an example, when the fault becomes serious, k_{fc} and φ_k decreases, as shown in Fig. 9.

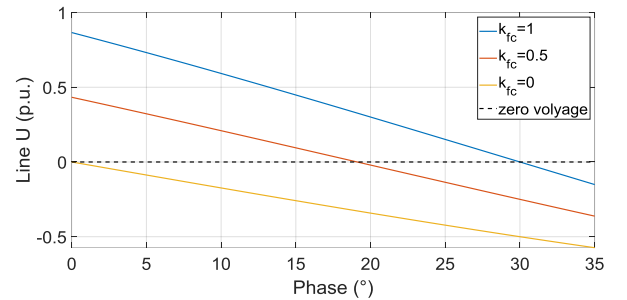


Fig. 9. The phase of the commutation voltage under SLG faults in varying degrees.

According to formula 2, i_{kf} can be expressed as

$$i_{kf} = C_f e^{\frac{-R_s t}{2L}} + I_f e^{\frac{-R_s t}{2L}} \quad 8$$

where C_f represents the constant term in the general solution. I_f can be written as

$$I_f = \int \frac{R_{sa} i_d + U_{\Delta f}}{2L} e^{\frac{R_s t}{2L}} dt \quad 9$$

By using Integration by parts, I_f can be expressed as

$$I_f = \left[\frac{R_{sa} i_d}{R_s} + A_7 \right] e^{\frac{R_s t}{2L}} \quad 10$$

where A_7 can be written as

$$A_7 = \frac{k_f U \sin(\omega t - \varphi_k - \varphi_R)}{\sqrt{R_s^2 + 4\omega^2 L^2}} \quad 11$$

where φ_R can be written as

$$\varphi_R = \arctan \frac{2\omega L}{R_s} \quad 12$$

where φ_R represents the phase change of the commutation voltage caused by SFCLs. C_f can be calculated as

$$C_f = \left[\frac{R_{sc} i_d}{R_s} - A_{12} \right] e^{\frac{R_s(\frac{\pi}{6} - \beta)}{2\omega L}} \quad 13$$

where A_{12} can be written as

$$A_{12} = \frac{k_f U \sin\left(\frac{\pi}{6} - \beta - \varphi_k - \varphi_R\right)}{\sqrt{R_s^2 + 4\omega^2 L^2}} \quad 14$$

This gives the expression for the phase change current.

The reduction of φ_k affects the selected value of the extinction angle γ_f .

$$\omega t|_{t=t_{ce}} = \varphi_k - \gamma_f \quad 15$$

The expression for γ_f under an AC short-circuit fault should be

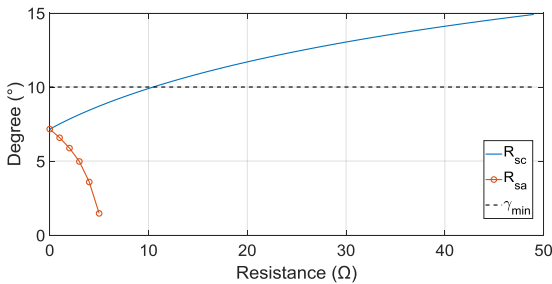
$$\left[\frac{R_{sc} i_d}{R_s} - A_{12} \right] e^{E_{15}} - A_{15} = -\frac{R_{sa} i_d}{R_s} \quad 16$$

where A_{15} and E_{15} can be written as

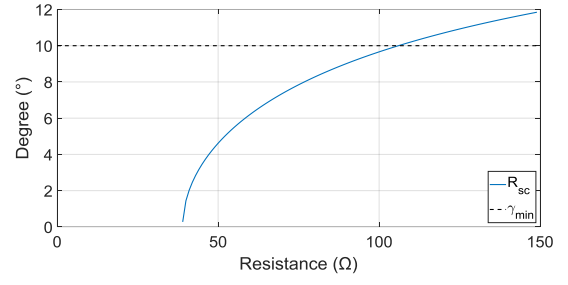
$$A_{15} = \frac{k_f U \sin(\gamma_f + \varphi_R)}{\sqrt{R_s^2 + 4\omega^2 L^2}} \quad 17$$

$$E_{15} = \frac{-R_s \left(\beta - \gamma_f + \varphi_k - \frac{\pi}{6} \right)}{2\omega L} \quad 18$$

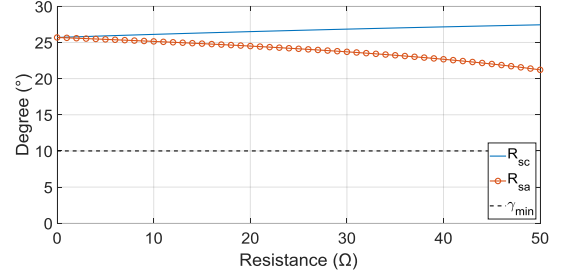
There is no analytical solution for γ_f . R_{sc} represents the resistance of SFCL on the outgoing phase. R_{sa} represents the resistance of SFCL on the incoming phase. It can be seen from formula 16 that they have different effects on γ_f during the commutation, which is shown in Fig. 10. The blue line indicates γ_f when the SFCL on the outgoing phase is inserted and the orange line with circles indicates γ_f when the SFCL on the incoming phase is inserted. Similarly, the fault on the outgoing or the incoming phase have different effects on the extinction angle. The fault on the outgoing phase will greatly reduce the extinction angle, while the fault on the incoming phase will increase the extinction angle. When the fault occurs on both the outgoing and the incoming phase, the extinction angle is reduced but the reduction is much smaller than that under the fault only on the outgoing phase. No matter what kind of fault occurs, the extinction angle is positively correlated with the resistance of SFCLs on the outgoing phase and negatively correlated with the resistance of SFCLs on the incoming phase. Therefore, during a commutation, it is only necessary to monitor whether the outgoing phase has a short-circuit fault.



(a)



(b)



(c)

Fig. 10. The extinction angle in varying limiting resistances during an AC short-circuit fault. (a) AC short-circuit faults occur on both outgoing and incoming phase; (b) AC short-circuit fault occurs on the outgoing phase; (c) AC short-circuit fault occurs on the incoming phase.

Properly increasing the inverted firing angle β or reducing i_d will also have the effect of increasing the extinction angle, which can reduce the requirement on the resistance of SFCLs. Take a fault only on the outgoing phase as an example, in order to make $\gamma_f = \gamma_{min}$, the requirement of R_{sc} under different β is shown in Fig. 11 and the requirement for R_{sc} under different i_d is shown in Fig. 12. It can be seen that increasing the inverted firing angle and limiting the DC current can greatly reduce the requirement for the resistance of SFCLs when the commutation failure is suppressed under the AC short-circuit fault.

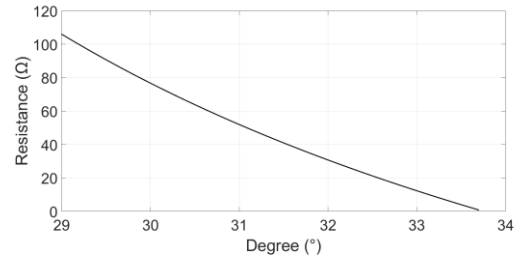


Fig. 11. The needed limiting resistances in varying inverted firing angles during an AC short-circuit fault on the outgoing phase.

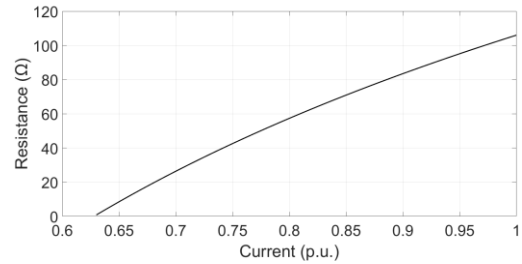


Fig. 12. The needed limiting resistances in varying DC currents during an AC short-circuit fault on the outgoing phase.

An SPG fault may also cause commutation failure. The fault will cause DC current on the non-fault pole to change: the current increases from the rated value then decrease to near the rated value when the short-circuit fault is eliminated or the fault current is limited. This can be equivalent to adding a current impulse i_{df} on the non-fault pole. The commutation circuit is shown in Fig. 13 and Fig. 14. L_{sd} and L_{sdn} represent the inductance of SFCLs on the fault pole and the non-fault pole. R_{sd} and R_{sdn} represent the resistance of SFCLs on the fault pole and the non-fault pole. i_{fr} and i_{fi} represent the fault current on the rectifier side and on the inverter side. L_d represents the smoothing reactor of the DC line. R_d represents the resistance of the DC line. k_{fd} represents the unit value of the DC voltage at the short-circuit fault point on the fault pole. Its valid range is $0 \leq k_{fd} \leq 1$. $k_{fd} = 0$ means zero-resistance grounding fault and $k_{fd} = 1$ means no short-circuit fault has occurred. U_d indicates the rated value of DC voltage. $-U_d$ indicates that the fault pole is negative. x represents the distance between the location of the short-circuit fault and the inverter as a percentage of the total length of the fault pole. M represents the mutual inductance of the bipolar DC system. t_0 represents the moment when the DC short-circuit fault occurs.

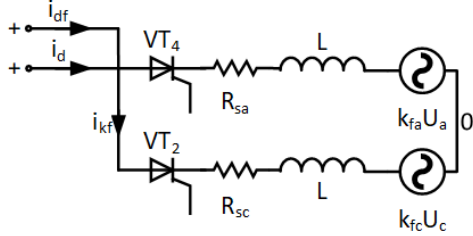


Fig. 13. The commutation circuit during an extra DC current impulse on the non-fault pole.

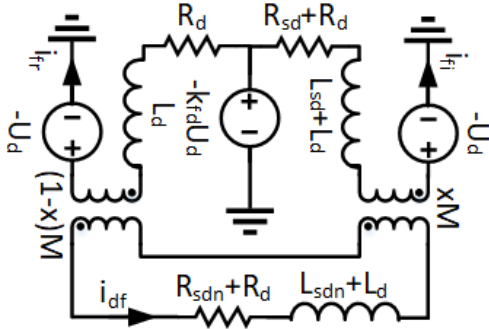


Fig. 14. An extra DC current on the non-fault pole.

After calculation, the expression of γ_f under an SPG fault can be obtained:

$$J_{171} - C_{176}e^{E_{174}} - A_{173} = \frac{R_{sa}i_d}{R_s} - J_{172} \quad 19$$

See the appendix for specific parameters.

It can be seen that the required resistance of SFCLs is related to the location of the short-circuit fault point, as shown in Fig. 15. The farther the short-circuit fault point from the inverter, the smaller the extra DC current on the non-fault pole and the smaller the resistance of SFCLs required to suppress the commutation failure.

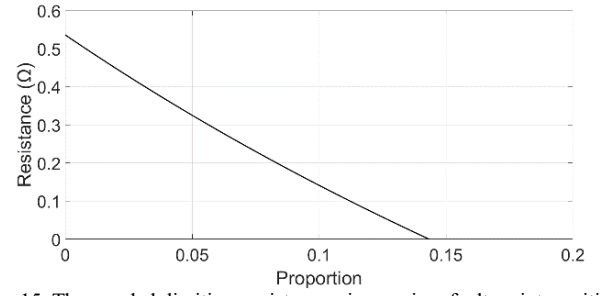


Fig. 15. The needed limiting resistances in varying fault point positions during a SPG fault.

The DC short-circuit fault current does not have periodicity, so t_0 will affect γ_f , as shown in Fig. 16. This is because the fault current costs time to grow and decay.

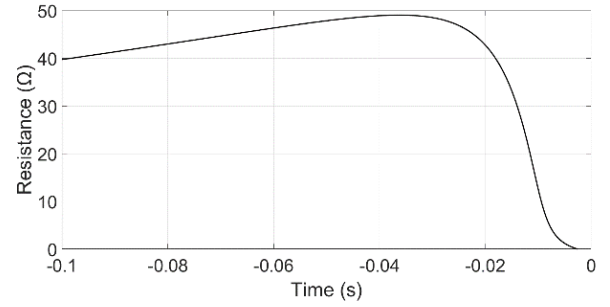


Fig. 16. The needed limiting resistances in varying DC short-circuit fault times during a SPG fault.

V. SIMULATION RESULTS

The proposed method is built in MATLAB/Simulink. The parameters of the double-pole DC transmission line is shown in TABLE I. The simulation model of SFCL coils is shown in [9].

TABLE I. THE PARAMETERS OF THE DOUBLE-POLE DC TRANSMISSION LINE

Rectifier side AC system	500 kV/60 Hz
Impedance on the rectifier side AC system	26.07 Ω/48.86 mH
Reactive power supply on the rectifier side AC system	600 Mvar
Inverter side AC system	345 kV/50 Hz
Impedance on the inverter side AC system	6.205 Ω/13.96 mH
Reactive power supply on the inverter side AC system	600 Mvar
Length of DC lines	300 km
Smoothing reactor	0.5 H
Fault duration	100 ms

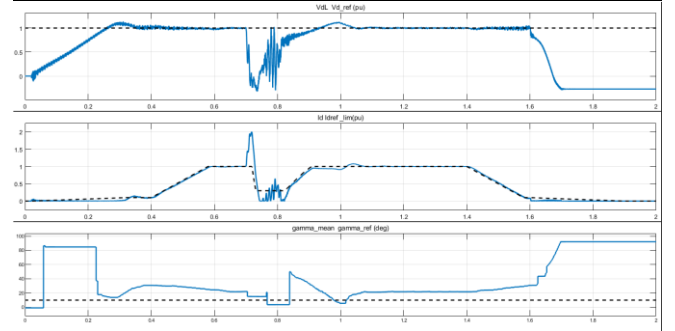


Fig. 17. The results of the double-pole DC line without the proposed method.

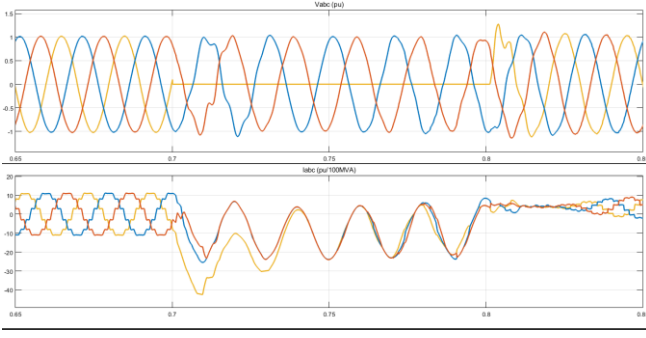


Fig. 18. The results of the AC line in the inverter side without the proposed method

Taking the fault grounding resistance of 0 and the fault start time of 0.7 s as an example, the model operation without the superconducting current limiting commutation converter strategy is shown in Fig. 17 and Fig. 18. From top to bottom, the blue solid line represents the actual values of DC voltage, DC current and inverter side turn-off angle respectively, and the black dotted line represents the given value and minimum turn-off angle of DC voltage and DC current respectively. Before the single-pole grounding short-circuit fault on the inverter side occurs, the bipolar DC transmission line has been started and operated normally. The DC voltage and current have reached the control target, and the turn-off angle is stable at 21.87° . After the fault occurs, the commutation failure suppression strategy of the inverter-side converter station starts, and the inverter angle increases by 23° , but it fails to avoid the occurrence of commutation failure. After the fault occurs, the voltage of the inverter-side fault phase drops to near 0, and the fault phase first undergoes commutation failure and causes DC short circuit. The DC current rises rapidly to 2 times the rated value, triggering the converter station to block. The excessive DC current further causes the commutation failure of the non-fault phase. The peak value of the fault current on the AC side reaches more than 4 times the rated value and the DC component of 2 times the rated value is introduced, and the AC current is completely out of step. The bipolar DC transmission line failed to restart many times until the unipolar ground short-circuit fault on the inverter side was removed.

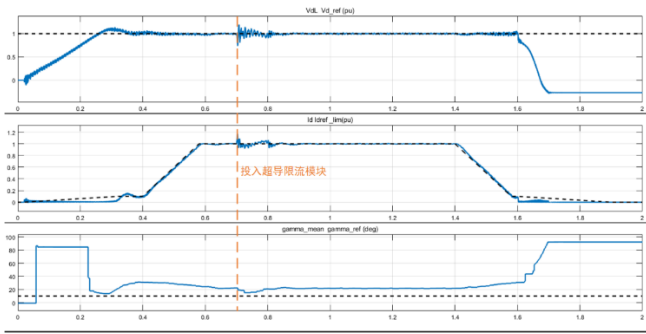


Fig. 19. The results of the double-pole DC transmission line with the proposed method

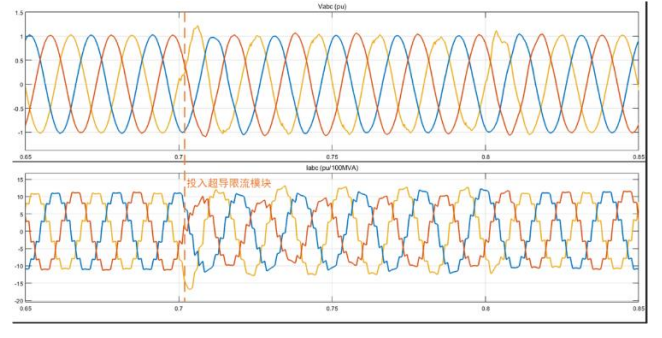


Fig. 20. The results of the AC line in the inverter side with the proposed method.

Under the same fault condition, the operation of the model after adopting the strategy proposed in this paper is shown in Fig. 19 and Fig. 20. It can be seen that after the fault occurs, the fault phase current increases, but the commutation process is still completed, and the DC short circuit is not caused. The inverter side can still maintain normal commutation during the fault. The slight increase of DC current at the moment of fault is caused by the increase of the AC current of the original phase. With the limitation of the fault phase current and the balance of the control strategy of the converter station, the DC current quickly returns to normal. During the continuous process of the fault, the DC voltage and DC current did not fluctuate violently, the power transmission was normal, and the blocking protection of the converter station was not triggered. In the case of the most serious fault with grounding resistance of 0, the commutation failure suppression strategy proposed in this paper is effective.

VI. CONCLUSION

This paper proposes a superconducting fault current limiting module topology and its control logic. Simulation models are built in MATLAB/Simulink to verify the effectiveness of the proposed method in suppressing commutation failures caused by serious faults.

The commutation with SFCLs under AC short-circuit fault and single-pole grounding short-circuit fault in the bipolar DC system is calculated in detail. The formulas for the extinction angle under different faults are obtained and analyzed. The influence of different parameters on the commutation and the resistance of superconducting coils.

The simulation results show that the proposed method can effectively suppress the first commutation failure. The proposed method can also effectively suppress subsequent commutation failures caused by thyristor valve controller faults. Its inhibition ability and effect are better than original strategies.

ACKNOWLEDGMENT

Project supported by the headquarters project of State Grid Corporation of China (5100-202199511A-0-5-ZN).

APPENDIX

In formula 19:

$$\begin{aligned}
R_{df} &= R_{sd} + R_d \\
L_{df} &= L_{sd} + L_d \\
R_{dfn} &= R_{sdn} + R_d \\
L_{dfn} &= L_{sdn} + L_d \\
D_d &= \frac{R_{sa}L_d - R_dL}{2R_dL - R_sL_d} \\
D_{df} &= \frac{R_{sa}L_{df} - R_{df}L}{2R_{df}L - R_sL_{df}} \\
D_{dfn} &= \frac{R_{sa}L_{dfn} - R_{dfn}L}{2R_{dfn}L - R_sL_{dfn}} \\
A_{171} &= \frac{1-x}{R_dL_{dfn} - R_{dfn}L_d} \\
A_{172} &= \frac{x}{R_{df}L_{dfn} - R_{dfn}L_{df}} \\
A_{173} &= \frac{k_f U \sin\left(\frac{\pi}{6} - \gamma_f - \varphi_k - \varphi_R\right)}{\sqrt{R_s^2 + 4\omega^2 L^2}} \\
E_{171} &= \frac{-R_d\left(\frac{\pi}{6} - \gamma_f - \omega t_0\right)}{\omega L_d} \\
E_{172} &= \frac{-R_{dfn}\left(\frac{\pi}{6} - \gamma_f - \omega t_0\right)}{\omega L_{dfn}} \\
E_{173} &= \frac{-R_{df}\left(\frac{\pi}{6} - \gamma_f - \omega t_0\right)}{\omega L_{df}} \\
E_{174} &= \frac{-R_s(\beta - \gamma_f)}{2\omega L} \\
C_{171} &= D_d e^{E_{171}} - D_{dfn} e^{E_{172}} \\
C_{172} &= D_{df} e^{E_{173}} - D_{dfn} e^{E_{172}} \\
C_{173} &= (D_{dfn} + 1)e^{E_{172}} - (D_d + 1)e^{E_{171}} \\
C_{174} &= (D_{dfn} + 1)e^{E_{172}} - (D_{df} + 1)e^{E_{173}} \\
C_{175} &= k_{fd} - 1 \\
C_{176} &= \frac{R_{sc}i_d}{R_s} - A_{12} \\
C_{177} &= A_{171}C_{171} - A_{172}C_{172} \\
C_{178} &= A_{171}C_{173} - A_{172}C_{174} \\
J_{171} &= MU_d C_{175} C_{177} \\
J_{172} &= MU_d C_{175} C_{178} e^{E_{174}}
\end{aligned}$$

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REFERENCES

- [1] Thio, C. V., & Davies, J. B. (1996). Commutation failures in hvdc transmission systems. *IEEE Transactions on Power Delivery*, 11(2), 946-957.
- [2] Rahimi, E. (2006). Commutation failure in single- and multi-infeed HVDC systems. *Iee International Conference on Ac & Dc Power Transmission*. IEEE Xplore.
- [3] Ruan S, Xu K, Liu D, et al. (2019) Statistical Analysis and Suggestions on Resistance Measures for Commutation Failure of HVDC Transmission System. *Automation of Electric Power Systems*, 43(18):13-17+34. (in Chinese)
- [4] Zheng, K., Shen, C., & Liu, F. (2018). Configuration scheme of statcom for mitigating simultaneous commutation failure risk of multi-infeed hvdc links. *Power System Technology*.
- [5] Yao W, Liu C, Fang J, et al. (2020). Probabilistic analysis of commutation failure in lcc-hvdc system considering the cfpv and the initial fault voltage angle. *IEEE Transactions on Power Delivery*, 35(2), 715-724.
- [6] Mirsacidi, S., Dong, X., Tzelepis, D., Said, D. M., Dysko, A., & Booth, C. (2018). A predictive control strategy for mitigation of commutation failure in lcc-based hvdc systems. *IEEE Transactions on Power Electronics*, 1-1.
- [7] Qiu, Q., Xiao, L., Zhang, Z., Jing, L., Liu, S., & Zhang, G. (2016). Investigation of flux-coupling type superconducting fault current limiter with multiple parallel branches. *IEEE Transactions on Applied Superconductivity*, 1-5.
- [8] Xiao, H., Li, Y., Gole, A. M., & Duan, X. (2019). Computationally efficient and accurate approach for commutation failure risk areas identification in multi-infeed lcc-hvdc systems. *IEEE Transactions on Power Electronics*, PP(99), 1-1.
- [9] Wang, H., Zhang, G., Zhang, Z., Liu, G., & Gao, H. (2020). A resistance model of fault current limiting coils under dc impact. *IEEE Transactions on Applied Superconductivity*, PP(99), 1-1.